

Transferring Shortcut Post-Training to Molecular and Protein Generative Models

From image flow matching to SE(3)-equivariant generation in structural biology

Andres Aranguren

Abstract

This document shows how a method built to speed up image generators can be carried over to models that generate proteins and molecules. The original method works like this: take a trained flow-matching model, freeze it, and add a small new input called the step size, which tells the model how big a jump to take. By training only this small part, the model learns to produce good results in just a few steps instead of many. The key reason this transfers is simple: the step size is just a number with no direction in space, so it can be fed in through a side channel without ever touching the 3D geometry the model works with. In protein and molecule models, which are carefully built so that rotating the input rotates the output, this means the method can teach the model to take big jumps without disturbing the physics it has already learned. We point to the specific models this applies to (FrameFlow, FoldFlow, and Proteina for protein backbones; EquiFM, MolFlow, and FlowDock for molecules and docking), the datasets that take the place of ImageNet and LAION-POP, the metrics that replace FID, and the one real change the new setting requires: following curved paths on the space of rotations instead of straight lines.

1 Background: the method being transferred

The starting point is a post-training recipe for flow-matching (FM) image generators. A standard FM model learns a velocity field $s_\theta(x_t, t)$ that transports noise $x_0 \sim \mathcal{N}(0, I)$ to data x_1 along the linear interpolant $x_t = (1 - t)x_0 + tx_1$. At inference one integrates the learned ODE, which typically requires tens to hundreds of function evaluations (NFE). The shortcut formulation of Frans et al. (2024) extends this by additionally conditioning the network on a *desired step size* d , so that a single model $s_\theta(x_t, t, d)$ can take jumps of size d and thereby generate under any inference budget, from many steps down to a single forward pass.

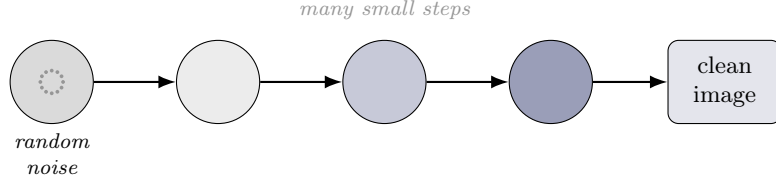
The original shortcut model trains this d -conditioned network from scratch. The method considered here instead *post-trains* an already-converged FM checkpoint: it injects the d pathway into the frozen model and trains only that pathway plus the conditioning projections, preserving the expensive pretraining investment while unlocking few-step sampling at a small additional cost. On DiT-S/8 over ImageNet this trains roughly one third of the parameters; on Flux.1-Dev (12B parameters) it turns a 32-step text-to-image model into a competitive few-step sampler in a short post-training run.

1.1 The conditioning pathway

The mechanism is worth stating precisely, because its structure is exactly what makes the method portable. The scalar d is mapped through a sinusoidal embedding followed by a two-layer MLP to a hidden-size vector **dte**, which is summed with the timestep embedding **te** and the class embedding **ye** to form a single conditioning vector

$$\mathbf{c} = \mathbf{te} + \mathbf{dte} + \mathbf{ye}.$$

Image generation



Protein generation

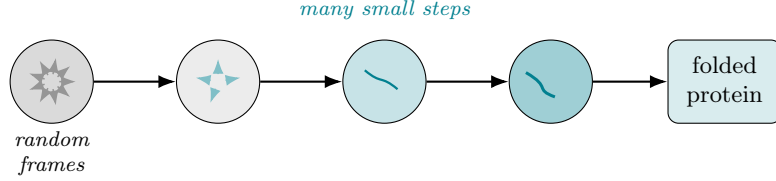


Figure 1: The two generation pipelines have the same shape. Both integrate an ODE from noise to a finished result over many small steps. The only difference is the object being transported: pixels (or VAE latents) in one case, residue frames (the positions and orientations of amino acids) in the other. The shortcut method reduces the step count in the top row; the goal of this work is to do the same for the bottom row.

Inside every transformer block, $\mathbf{c}$ is projected by a block-specific matrix W_{ada} and split into six control signals: shift, scale, and gate for the attention sublayer, and the same three for the MLP sublayer. These signals modulate the data stream through adaptive layer normalisation (adaLN-Zero). Crucially, after this projection $\mathbf{c}$ *disappears*: the patch-token stream $\mathbf{x}$ flows through layer normalisation, the attention projections W_Q, W_K, W_V, W_O , and the MLP, and *never receives* $\mathbf{c}$ *directly*. The conditioning touches the data only through the shift, scale, and gate values.

1.2 Why the model starts identical to its teacher

The final layer of the d embedder is zero-initialised, so that $\mathbf{dte} = 0$ for every d at the start of post-training. The conditioning vector therefore collapses to $\mathbf{c} = \mathbf{te} + \mathbf{ye}$, exactly the distribution the pretrained projections were trained on, and the post-trained model is bit-for-bit identical to the teacher at step zero. The d pathway then learns useful structure gradually. The loss combines an FM anchor term that holds the model at its pretrained quality with a self-consistency term that bootstraps many-step behaviour into few-step behaviour:

$$\mathcal{L}(\theta) = \underbrace{\mathbb{E}[\|s_\theta(x_t, t, 0) - (x_1 - x_0)\|^2]}_{\mathcal{L}_{\text{FM}}} + \underbrace{\mathbb{E}[\|s_\theta(x_t, t, 2d) - s_{\text{target}}\|^2]}_{\mathcal{L}_{\text{SC}}},$$

$$s_{\text{target}} = \frac{1}{2}(s_{\theta_{\text{EMA}}}(x_t, t, d) + s_{\theta_{\text{EMA}}}(x'_{t+d}, t+d, d)), \quad x'_{t+d} = x_t + \frac{d}{2} s_{\theta_{\text{EMA}}}(x_t, t, d).$$

The self-consistency target is produced by an exponential-moving-average (EMA) teacher under stop-gradient, which damps the oscillations that a self-referential objective would otherwise induce. Only the d embedder and the W_{ada} projections receive gradients; everything else is frozen.

2 Why the field is the right place to apply this

Generative modelling in structural biology runs on exactly this engine. The dominant protein backbone generators are flow-matching models over the manifold of residue frames: a protein of N residues is represented as a sequence of rigid-body transformations $T \in \text{SE}(3)^N$, following

the frame parametrisation introduced by AlphaFold2. FrameFlow adapts the FrameDiff diffusion model to flow matching and reports roughly five times fewer sampling steps at better designability; FoldFlow builds SE(3) (and stochastic SE(3)) flow matching with optimal-transport couplings; and Proteina scales flow-based backbone generation to large models and datasets. P2DFlow applies SE(3) flow matching to conformational ensembles. On the small-molecule side, equivariant flow-matching models such as EquiFM and MolFlow generate 3D conformers, FlowDock learns a geometric flow over ligand poses for blind docking, and discrete flow models extend the same framework to coupled atom-type and coordinate generation. Materials-structure analogues (MolCrystalFlow, MOFFlow) use the identical machinery.

The practical bottleneck across all of these is sampling speed. Diffusion- and flow-based structural generators typically require hundreds of iterative steps, and de novo design needs thousands to millions of candidate structures to find a few that fold and function. This makes inference cost the binding constraint, precisely the constraint the shortcut post-training method was built to relax. The problem is recognised in the field: *Distilled Protein Backbone Generation* (Xie et al., 2025) adapts score-identity distillation to protein backbones and reports more than a twenty-fold sampling speedup at comparable designability to a Proteina teacher, while noting that a naive transfer of image distillation methods produces unacceptably low designability and that careful, domain-aware adaptation is required. That paper validates both the demand for few-step structural generators and the need for an adaptation that respects the domain, which is what the present method offers from a different, post-training-based angle.

3 The portability argument

The reason the recipe transfers is not superficial similarity of architectures; it is a property of what d is. Conditioning signals in an SE(3)-equivariant network fall into two categories. The first is geometric, equivariant data: the residue frames, atomic coordinates, and pair representations. When the input is rotated or translated, these quantities transform accordingly, and the network’s update rules are constructed (through invariant point attention, tensor-product layers, or equivariant message passing) to respect that transformation. The second category is scalar invariants: the diffusion or flow time t , an optional fold-class or property label, and, in the transferred method, the step size d . These carry no geometric type; they are unchanged under any rotation or translation of the structure.

A scalar invariant can only be injected through invariant channels. In a frame-based network this is the adaLN-style modulation of the invariant feature track, or an additive bias inside invariant point attention; in an equivariant message-passing network it is the modulation of the scalar (type-0) features. Because d enters only these invariant pathways and never alters the equivariant update rules, adding the d pathway *cannot break the network’s equivariance*. This is the molecular analogue of the clean $\mathbf{c}/\mathbf{x}$ boundary in the image model: there, $\mathbf{c}$ touched only W_{ada} and never the data stream; here, d touches only the invariant conditioning and never the equivariant geometry. The portability of the method is therefore a direct consequence of d being a scalar, not a fortunate empirical coincidence.

4 What stays the same and what changes

4.1 Carried over unchanged

The two-term loss, the EMA teacher under stop-gradient, the freezing of the backbone with training restricted to the conditioning pathway, the zero-initialisation that yields identity-at-start, and the sinusoidal-plus-MLP embedding of the scalar d all carry over verbatim. These are the parts of the method that depend only on the existence of a conditioning vector and a frozen teacher, both of which the target models have.

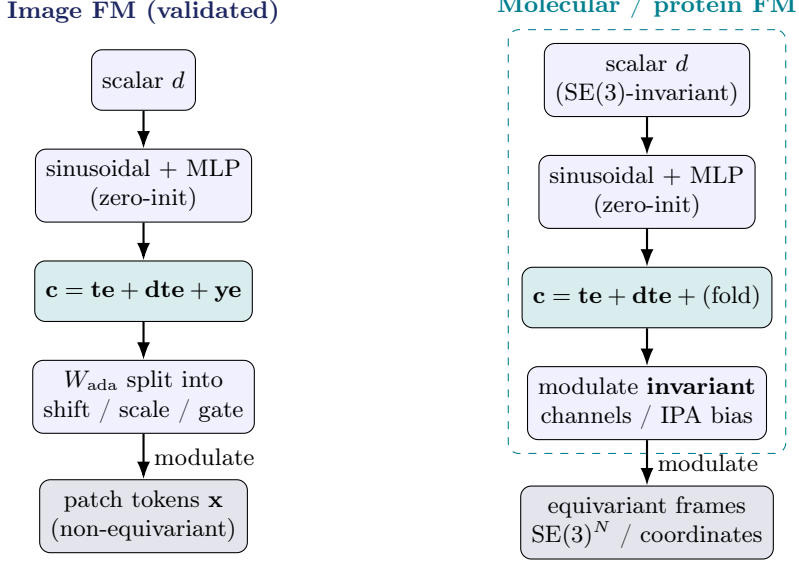


Figure 2: The conditioning pathway is structurally identical across domains. The step size enters through a side channel, becomes part of the conditioning vector, and only sets the adaptive-normalisation modulation. It never touches the data stream. In the molecular case the modulated stream is equivariant, but because d is a scalar invariant it enters through invariant gates and the equivariance is preserved by construction.

4.2 Changed by the manifold structure

The substantive change is geometric. The image method takes a half-step by vector addition, $x'_{t+d} = x_t + \frac{d}{2} s_{\theta_{\text{EMA}}}(x_t, t, d)$, and averages two velocities by ordinary addition. On a curved manifold these operations must respect the geometry. The half-step becomes an exponential map, and velocities are tangent vectors that must be parallel-transported to a common tangent space before averaging:

$$x'_{t+d} = \exp_{x_t} \left(\frac{d}{2} v_{\theta_{\text{EMA}}}(x_t, t, d) \right), \quad v_{\text{target}} = \frac{1}{2} \left(v_{\theta_{\text{EMA}}}(x_t, t, d) \oplus \text{PT}[v_{\theta_{\text{EMA}}}(x'_{t+d}, t+d, d)] \right),$$

where $\text{PT}[\cdot]$ denotes parallel transport of the second tangent vector back to $\mathcal{T}_{x_t}\mathcal{M}$. For the frame manifold $\text{SE}(3)^N$ this is less daunting than it appears: the translation component is ordinary Euclidean space, so the original formula is unchanged there, and only the rotational component on $\text{SO}(3)$ requires the exponential and logarithm maps, which the target codebases already implement for their FM training. The adaptation is therefore localised to the half-step and averaging operations, not a rewrite of the method.

A second change is the conditioning injection site. Frame-based networks do not apply a plain layer normalisation to coordinates, so the adaLN-Zero modulation maps onto whatever invariant modulation the target architecture provides, typically a bias or scale inside invariant point attention, or modulation of the scalar feature track. The principle is identical; the projection target differs.

The third change is evaluation, discussed below.

5 Concrete target 1: protein backbone generation

The natural first target is protein backbone generation, for several reasons. Open pretrained checkpoints and training code are available (FrameFlow at [microsoft/protein-frame-flow](https://github.com/microsoft/protein-frame-flow), FoldFlow at [DreamFold/FoldFlow](https://github.com/DreamFold/FoldFlow)), the frame representation already separates invariant scalars from

equivariant frames cleanly, the backbone-only setting keeps dimensionality modest enough for single-node post-training, and designability has a crisp automatable definition.

Teacher and injection. Take a converged SE(3) FM backbone generator as the teacher, FrameFlow for fast iteration, or Proteina for a stronger baseline. Embed the scalar d through a sinusoidal-plus-MLP block with a zero-initialised final layer, sum it into the existing time and fold-class conditioning, and train only the d embedder and the invariant modulation projections while freezing invariant point attention, the triangle layers, and all other backbone weights.

Data. The training distribution plays the role ImageNet and LAION-POP played in the image work. The standard choice is PDB monomers, length-filtered (for example 60 to 256 residues) and clustered at 30 to 40% sequence identity to remove redundancy, the same split FrameFlow and FoldFlow train on. SCOPe or CATH fold labels provide the optional class conditioner that the ImageNet class label provided. For scaling, an AlphaFold-DB distillation set mirrors how Proteina scales up. As in the Flux pipeline, frames and conditioner embeddings are pre-cached so that no auxiliary encoder runs during post-training.

Evaluation. FID is meaningless for structures and is replaced by a self-consistency pipeline. *Designability* is the fraction of generated backbones with self-consistency RMSD below 2 Å: each generated backbone is passed to ProteinMPNN to design a sequence, the sequence is folded by ESMFold, and the refolded structure is compared by RMSD to the generated backbone. *Diversity* counts structural clusters (via MaxCluster or Foldseek) among designable samples, and *novelty* is the maximum TM-score against the PDB, with lower meaning more novel. The headline study mirrors the image work exactly: plot designability, diversity, and novelty against NFE, and against post-training steps, and show that designability is held at the teacher’s level while the step count collapses from roughly one hundred or more down to a handful, training under forty percent of the parameters.

6 Concrete target 2: small molecules, conformers, and docking

The same recipe applies to molecular generation with different data and metrics. For 3D conformer generation, the teacher is an equivariant FM model in the EquiFM or MolFlow family, trained on GEOM-Drugs and QM9; the relevant metrics are coverage and matching RMSD against reference ensembles, together with chemical validity and strain energy, each reported against NFE. For structure-based drug design, the teacher generates ligands inside a protein pocket (as in the MolFORM family), trained on CrossDocked2020 or PDBbind, with the pocket embedding serving as the cached conditioner; the metrics are docking score (for example AutoDock Vina), drug-likeness (QED), synthetic accessibility, and pose RMSD. Docking itself is a high-value target: FlowDock’s geometric flow over ligand poses, reduced to a few steps, would raise virtual-screening throughput substantially, since screening is dominated by the per-pose sampling cost.

7 Domain-specific risks

Three risks are specific to the molecular domain, and the recipe already contains the means to address the most important one.

One bad step ruins the structure. In image generation a small error in a few-step jump produces a slightly degraded but still plausible image. In structural biology a single bad jump can produce an undesignable backbone or an invalid molecule with broken bonds or steric clashes. The mitigation is built into the method: the $\mathcal{L}_{\text{FM}}$ anchor and the EMA stop-gradient teacher

keep the few-step student close to the validated teacher, and evaluation reports designability and validity rather than only a distributional score, so failures are visible rather than hidden in an average.

Manifold geometry. The self-consistency half-step must respect $SO(3)$ geometry. As noted, this is a localised change confined to the exponential and logarithm maps on the rotational component; the translational component is unchanged from the Euclidean formula.

Stochastic teachers. If the teacher is a stochastic $SE(3)$ flow (FoldFlow-SFM), the deterministic shortcut should be distilled against the teacher’s probability-flow ODE rather than its SDE, and any inference-time annealing the teacher relies on must be accounted for in the targets.

8 Summary

Structural-biology generation is dominated by $SE(3)$ flow-matching models that are too slow for the candidate volumes real design requires, the same bottleneck the shortcut method addressed for images. The method transfers because the step size d is a scalar invariant: injecting it through adaptive-normalisation invariant gates leaves the equivariant backbone, and the physics it encodes, untouched. The procedure is otherwise unchanged: freeze the backbone, train the d embedder and the conditioning projections, zero-initialise for identity-at-start, and keep the $\mathcal{L}_{FM} + \mathcal{L}_{SC}$ loss with an EMA teacher. Only two things change: the half-steps become geodesic on $SO(3)$, and designability and validity metrics replace FID. The first milestone is FrameFlow on PDB, with designability plotted against NFE and against post-training steps, the identical two-axis study already run on DiT-S/8 and Flux. The thesis is that a pretrained, expensive, many-step structural generator can be turned into a few-step sampler by training a small invariant conditioning pathway, almost for free, without retraining and without disturbing the equivariant physics the model has already learned.

Key references

- Frans et al., *One Step Diffusion via Shortcut Models*, 2024.
- Yim et al., *Fast protein backbone generation with $SE(3)$ flow matching* (FrameFlow), 2023.
- Bose et al., *$SE(3)$ -Stochastic Flow Matching for Protein Backbone Generation* (FoldFlow), 2024.
- Geffner et al., *Proteina: Scaling flow-based protein structure generative models*, ICLR 2025.
- Jin et al., *P2DFlow: A Protein Ensemble Generative Model with $SE(3)$ Flow Matching*, JCTC 2025.
- Xie et al., *Distilled Protein Backbone Generation*, 2025.
- Lipman et al., *Flow Matching for Generative Modeling*, 2022.
- Black Forest Labs, *FLUX.1*, 2024; Cai et al., *Shortcutting Pre-trained Flow Matching Diffusion Models*, 2025.